

Analysis of Data Anonymization Techniques

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Abstract: The privacy of personal data is a very important issue these days. How to process the data and use it for analysis without compromising the individual's identity is a critical task and must be done in order to ensure the anonymity of this data. To try to unanimously unify this anonymity, laws and regulations such as GDPR were created. In this paper, GDPR will be described and the concepts of anonymization and pseudonymization will be explained. We present some of the main anonymization techniques and efficient software to support the application of these techniques. The main objective is to understand which techniques offer a higher level of anonymization, the strengths and weakness of each one and the advantages in its use.

1 INTRODUCTION

In recent years, the exponential growth of digital information has increased more than expected with technological development. Nowadays, about 53% (ONU News, 2019) of the world population has access to Internet and, for this reason, their personal data is spread out and accessible. Now, a technological area in great expansion, precisely due to the large volume of data available, is the area of data analysis.

What happens in large organizations and which has also been growing in smaller ones is that they carry out analyzes on their data in order to find patterns, trends and customer profiles. This would not be a problem if these data were processed internally within the organization. But what is happening is that a lot of data is sold to other organizations or made available to the public for research purposes as a consultation service. These personal data may contain information that allows to identify the individual and being made public may violate privacy. With this, a very debated question has been how to maintain the privacy of this data without rendering it useless for analysis.

To answer this question, data protection laws and regulations have emerged around the world, such as

GDPR (General Data Protection Regulation, s.d.), and with them the concept of data anonymization that allows the removal of personal identity from data through anonymization techniques. But can these techniques be reversible? Will it be possible to identify the individual again after its application? What data should or should not be anonymized? These are some of the questions that arise in this research topic.

In this paper, the concepts involved in data anonymization will be addressed, some of the techniques used for this anonymization will be studied, which are the risks of re-identification associated and the analysis of some software tools that allows to perform these techniques.

The rest of this paper is organized as follows. Section 2 presents the concepts of GDPR, anonymization and pseudonymization. In Section 3, the main techniques used for anonymizing data are presented. Section 4 report the risks associated with the re-identification of each technique. Then, in Section 5, some software is presented that allows applying the techniques described in section 3. Finally, section presents the conclusions of the study and future work.

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2 DEFINITIONS AND CONCEPTS

To better understand the framing of this topic, it is necessary to clarify some of the concepts. In this section, we give an understanding what the GDPR consists of and what it means to anonymize data.

2.1 GDPR

The GDPR – General Data Protection Regulation – is an official European regulation that aims to harmonize privacy and data protection laws in all member states. It has been applied since May 25th, 2018 (General Data Protection Regulation, s.d.). In general, this regulation contains clauses and requirements on how personal data and information are treated and is applicable to all companies operating in the European Economic Area, regardless of their country of origin. It is also required that the stored data must be anonymized or pseudonymised in such a way that they do not allow to identify any individual again.

2.2 Pseudonymization vs. Anonymization

The GDPR has many requirements about how data is treated and differentiates personal data from anonymized and pseudonymised data.

Therefore, according to article 4 of the GDPR:

- Personal data is “any information relating to an identified or identifiable natural person (*data subject*)”;
- Pseudonymised data is processed personal data that can no longer be attributed to a specific holder without the use of additional information;
- Anonymized data is personal data treated in such a way that it is impossible to re-identify or deduce information about a specific individual.

So, if there is any way to identify the individual data subject, the data were not anonymous but pseudonymized.

Anonymization is important because these data can be used for analysis by companies and does not offer great risks to the data subject because, according to Recital 26, “the principles of data protection should therefore not apply to anonymous information, namely information which does not relate to an identified or identifiable natural person or to personal data rendered anonymous in such a manner that the data subject is not or no longer identifiable.” That is, when a dataset is correctly anonymized, GDPR is no longer applicable.

Thus, it is important to know the data anonymization techniques and how they can be applied.

3 DATA ANONYMIZATION TECHNIQUES

In order to correctly choose the anonymization techniques to be applied, we must understand what is the purpose of this anonymization because the different techniques have different characteristics and may be more or less suitable for certain purposes. The three most used ways to change data are to replace, modify or remove an attribute or a record.

Note that it is important to be able to maintain the usefulness of the data and at the same time respect the privacy terms.

In this section, we describe some of the data anonymization techniques and in what situations they should be applied.

3.1 Remove Attributes (Suppression)

In this technique, an attribute is removed from the dataset. This should happen whenever an attribute is not relevant or necessary for analysis or whenever it is impossible to anonymize it in any other way. In the example given in the Guide to Basic Data Anonymization Techniques (Personal Data Protection Commission Singapore, 2018) for this technique, in which it was intended to analyse students' grades in an assessment test, the dataset was composed of three attributes: student name, trainer and grade. Figure 1 shows an example of the original dataset.

Before anonymisation:

Student	Trainer	Test Score
John	Anna	93
Nicholas	Paul	86
Josh	Paul	54
Taylor	Anna	78

Figure 1: Suppression – Original dataset.

In order to anonymize the data, the attribute “student name” was removed using the technique of removing attributes, as we can see in the figure below.

After suppressing the "student" attribute:

Trainer	Test Score
Anna	93
Paul	86
Paul	54
Anna	78

Figure 2: Suppression – Anonymized dataset.

Suppression can also occur for a complete dataset record affecting several attributes.

The main advantage of this technique is that, when permanently deleting an attribute or record, it becomes impossible to retrieve the information.

3.2 Character Replacement

The substitution of characters consists of covering up characters of an attribute or value of the data by replacing those characters with a predefined symbol (for example, by X or *). This substitution can be partial, partially hiding a text or attribute, which may be sufficient to anonymize its content.

Also, in the Guide to Basic Data Anonymization Techniques (Personal Data Protection Commission Singapore, 2018), an example can be found in which to make an analysis of a dataset where the post code was identified, the last 4 digits of the post code were replaced by the character 'X'. Figure 3 shows the original dataset and the anonymized dataset.

Before anonymisation:

Postal Code	Average No. of Orders/month
100111	2
200222	8
300333	1

After suppressing the "student" attribute:

Postal Code	Average No. of Orders/month
10xxxx	2
20xxxx	8
30xxxx	1

Figure 3: Example of replacing characters.

3.3 Shuffling

In this technique, the data is randomly mixed or reorganized and the values of the original attributes remain in the dataset but can be associated with another record. This technique can be used when it is

intended to analyse only one attribute and it is not necessary to relate it to the others.

For example, if we want to analyse the amount of sales in a given region, it is only necessary to use the attribute 'region' and the permutation does not influence the results because a certain region will occur the same number of times before and after the permutation.

However, this technique does not always provide anonymization of the data and it may be possible to reorganize it to its original form. Therefore, it must be used in conjunction with other techniques.

3.4 Noise Addition

The addition of noise is one of the most used data anonymization techniques, being applied by several technological "giants" such as Google (Google, s.d.). The technique is to slightly modify the attributes of the dataset making them less accurate. An application of this technique would be, for example, to add or subtract days or months to a date.

Although this technique allows to hide the real values, it is necessary to understand the level of noise that must be applied in order to have little impact on data analysis and individuals' privacy.

3.5 Generalization

Generalization is another approach used by Google (Google, s.d.) and consists of generalizing the attributes in order to change the respective scale or order of magnitude.

An example of this is to replace the "date" attribute (day/month/year) with the "year" attribute, removing the day and month.

Like the addition of noise, this approach may prevent the individual from being identified, but it may not result in effective anonymization.

There are two techniques that can be considered generalization: K-Anonymity and L-Diversity, described in the following subsections.

3.5.1 K-Anonymity

This technique consists of grouping the records of K individuals into categories making them fall under the same combinations. Thus, each record in the dataset "is similar to at least K-1 other records" (El Emam, 2008).

For example, if the identifying attributes are age and disease and K=3, the dataset anonymized by this method will have at least 3 records for each combination of the identifying attributes. Considering the two individuals in the example illustrated in

Figure 4, the result of k-anonymity with $K=3$ would be the one illustrated in Figure 5.

Paul	
Age	Disease
21	Heart Disease

Mark	
Age	Disease
38	Cancer

Figure 4: K-Anonymity: original dataset.

Age range	Disease
20-30	Heart Disease
20-30	Heart Disease
20-30	Heart Disease
30-40	Cancer
30-40	Cancer
30-40	Cancer

Figure 5: K-Anonymity: original dataset.

After the application of K-Anonymity, the probability of identifying an individual is equal to or less than $1/K$. Therefore, the higher the K , the lower the probability of identification.

3.5.2 L-Diversity

Based on Machanavajjhala et al. (2007), L-Diversity is an evolution of K-Anonymity in which at least L distinct values must exist for each equivalent group and sensitive attribute (identifier). That is, it is guaranteed that in each equivalent group each attribute has at least L different values.

The objective of this technique is to limit the occurrence of equivalence classes with low variability of the attribute. Thus, an intruder who has access to data for specific individual always remains with a degree of uncertainty.

However, this technique is susceptible to attacks of probabilistic inference.

4 RE-IDENTIFICATION RISKS

In anonymization processes where it is crucial to maintain data privacy, there is always a need to assess the risks of re-identification.

There are three types of risks that should be managed (CHEO Research Institute, s.d.):

- *Prosecutor Risk*: re-identifying a record knowing that the individual exists in the dataset;
- *Journalist Risk*: re-identifying a record without being sure that the individual exists in the dataset;
- *Marketer Risk*: re-identifying large volumes of records.

When the risk of re-identification is measured, the following conditions are valid (CHEO Research Institute, s.d.):

- The prosecutor risk will be equal to or larger than journalist risk;
- Journalist risk will be equal to or larger than marketer risk;
- Prosecutor risk will be equal to or larger than marketer risk.

Risk estimates are calculated assuming that a potential attacker has access to only the dataset available. These results are usually presented for each type of risk using indicators that show the percentage of records at risk, highest risk and success rate. In the case of marketer risk, only the success rate is calculated.

Each company is responsible for establishing the risk limit for re-identification that it is willing to accept and is responsible for assessing the risk associated with the processing of personal data.

In the Table 1, the anonymization techniques presented above are identified and those that allow or can easily allow re-identifying an individual through the dataset.

Table 1: Risk of re-identification for each technique.

Technique	Allows re-identification?
Suppression	No
Character replacement	Yes
Shuffling	Yes
Noise Addition	Yes
K-Anonymity	No (minimum)
L-Diversity	No

In the next section, some software is presented for the application of anonymization techniques and that perform the risk analysis.

5 DATA ANONYMIZATION TOOLS

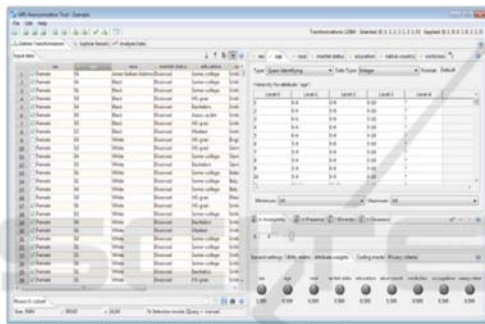
As mentioned earlier, there is anonymization software that allows to systematically apply the techniques described.

Based on the work developed in 2014 by Bergeat et al. (Maxime Bergeat, 2014) and in 2017 by Pinho (Pinho, 2017), software for these purposes are presented below.

5.1 ARX

ARX is an open source framework developed in Java (ARX, s.d.). It allows to implement several of the techniques described, such as K-Anonymity and L-Diversity, and also to implement a set of metrics to assess the loss of information.

This software has a graphical tool with a simple and intuitive interface, shown in Figure 6, which supports the import and cleaning of data, wizards for creating transformation rules, intuitive ways to adapt the anonymized dataset to the requirements and visualizations of risks and re-identification.



Source: <https://arx.deidentifier.org/overview/>

Figure 6: ARX Interface.

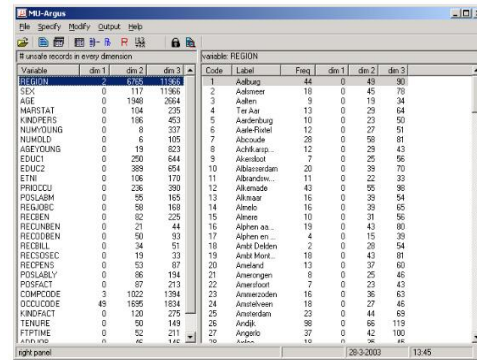
ARX is also available as a library with an API that provides data anonymization capabilities for any Java program. ARX is compatible with SQL databases, Microsoft Excel and CSV files.

Succinctly, to perform anonymization, we start by importing the dataset, defining the indicators and sensitive variables and defining the generalization hierarchies. At the end, the transformations are extracted.

5.2 μ -Argus

Argus stands for ‘Anti Re-identification General Utility System’ and is a software developed to create safe microdata files (MU-Argus, s.d.). It is based on R programming language and uses different statistical methods of anonymization such as noise addition and suppression. It can also be used to generate synthetic data.

Its interface can be seen in Figure 7.



Source: <https://aircloak.com/top-5-free-data-anonymization-tools/>

Figure 7: μ -Argus Interface.

Anonymization steps involve defining sensitive indicators and variables, estimating the risk of disclosure and re-identification, implementing methods that reduce this risk and exporting microdata.

5.3 SDCMicro

SDCMicro is a free open source package for researchers and public use (International Household Survey Network, s.d.). Can be used to generate anonymous data by creating files that can be used by the public and scientific researchers.

This tool allows to apply various anonymization techniques such as suppression, adding noise and shuffling and includes functions to measure the risk throughout the process.

It is provided as a user-friendly GUI in which users unfamiliar with R can implement anonymization methods. Supports the import and export of microdata in various formats such as STATA, SAS, SPSS, CSV and R.

Figure 8 shows its interface.



Source: <https://ihnsn.org/software/disclosure-control-toolbox>

Figure 8: SDCMicro Interface.

One of the advantages of this tool is that it includes functions to measure, visualize and compare

risk and utility during the anonymization process, helping organizations to prepare reports.

5.4 Privacy Analytics Eclipse

According to the website itself (Privacy Analytics, s.d.), this tool anonymizes data allowing it to maintain its quality and preserve compliance with many data privacy regulations, including GDPR. It also allows to adopt HIPPA's Expert Determination Method that classifies data attributes. It works with large volumes of data and, like the previous ones, offers re-identification risk assessment. Supports data export in CSV and ODS formats.

This tool is widely used in the healthcare area. Its main advantage is that it is a fast and very precise anonymization tool that guarantees compliance with legal regulations. Anonymization techniques are optimized based on measures of risk to patient privacy.

5.5 Software vs Techniques

Table 2 shows the software listed above and which is usually used for the application of the studied techniques.

Table 2: Software vs Techniques.

Software/Tool	Techniques
ARX	Generalization K-Anonymity L-Diversity Suppression
μ -Argus	Noise Addition; Suppression
SDCMicro	Noise Addition Suppression Shuffling
Privacy Analytics Eclipse	Generalization K-Anonymity L-Diversity Noise Addition Shuffling

6 CONCLUSIONS AND FUTURE WORK

Anonymization is an important issue that has been increasingly demanding the attention of the community. With the large volume of personal data available for analysis and treatment there is a need to ensure the privacy of individuals.

If, on the one hand, GDPR harmonises the level of data protection, on the other hand, the fact that there are defined rules, allows companies to carry out more actions with the information, allowing them to analyse and adopt the information to assist business decisions.

There are several anonymization techniques, the main ones being presented in this paper. Each technique has advantages and weaknesses; however, it is necessary to choose the appropriate technique for the dataset to be worked on at the moment. Therefore, anonymization techniques guarantee data privacy when properly applied. In some specific situations, it may be advantageous to apply several combined techniques. In many cases, after applying anonymization techniques to the dataset, it may be possible, in some way, to infer information about an individual, even if is not very accurate.

The need to implement and comply with the defined standards, as in the GDPR, means that there are several tools and software capable of assisting the anonymization of data, in addition to those presented in this article. For example, in (Privacy Analytics Eclipse Alternatives & Competitors, s.d.) is a list of 20 alternative tools to Privacy Analytics Eclipse. In general, all of them allow to apply more than one anonymization technique and include features of risk assessment of re-identification. Note that some of them are specific to a purpose or to work with a certain type of data.

As future work, we intend to test each of these tools with real datasets and evaluate the anonymization performance of each one.

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